

Applications

It is the applications that make Google Android really unique. Android ships with a host of applications including an email client, SMS program, calendar, maps, browser, contacts, and others. All these applications are written in Java and run with their own instances of Dalvik virtual machine. Dalvik has been written so that a device can run multiple VMs efficiently. The Dalvik VM executes files in the DEX format which is optimised for minimal memory footprint. The VM is register-based (unlike stack based), and runs classes compiled by a Java language compiler that have been transformed into the DEX format. The Dalvik executables are then modified by the OS again when they get installed on to the mobile device to gain further optimisations, byte order gets swapped in certain data, simple data structures and function libraries are linked inline, and empty class objects are short-circuited to give a much more efficient program run time. The Dalvik VM relies on the Linux kernel for underlying functionality such as threading and low-level memory management. Using Dalvik system makes running and management of processes on a mobile device much more efficient.



Default Homescreen of iPhone 3GS running OS 3.0

One great asset of this OS is the integration with basic Google applications which is excellent. If your primary email client is Gmail, the experience on this OS very similar to how it is on a desktop browser. Gmail, Google Calendar and your contacts can be set to sync automatically so your device is constantly updated.

Home

The starting interface of the Android OS is vivid & beautiful and involves a Home screen which can be populated with shortcuts, widgets, a notifications bar and a main applications menu. The Home screen can be spanned into different screens (3 for G1) which you can swipe left or right to find additional free space to put in shortcuts. The applications menu is brought up by tapping on an icon at the bottom of the screen. Pressing the Menu button will bring up a context menu relevant to the application you have opened at the time. The notifications bar appears when you put your finger